

Isometrical embeddings of ultrametric spaces into non-archimedean valued fields

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In this paper we present a technique yielding the following theorems.

THEOREM 1. *For each ultrametric space X there exists a non-archimedean valued field K such that X can isometrically be embedded into K .*

THEOREM 2. *The following conditions for a non-archimedean non-trivially valued complete field K are equivalent.*

- (α) *K is spherically (= maximally) complete. Its residue class field is finite.*
- (β) *Each K -valued isometry defined on a subset of K can be extended to an isometry $K \rightarrow K$.*
- (γ) *Each isometry $K \rightarrow K$ is surjective.*

For definitions, terminology, notations we refer to [1]. The next lemma contains the essentials.

KEY LEMMA. *Let (X, d) be an ultrametric space and let $(K, |\cdot|)$ be a spherically complete non-archimedean valued field such that*

- (i) $\{d(x, y) : x, y \in X\} \subset \{|\lambda| : \lambda \in K\}$,
- (ii) *if $A \subset X$ is a set of equidistant points, $a \in A$, then the cardinality of $A \setminus \{a\}$ is strictly less than the cardinality of the residue class field k of K .*

Let Y be a subset of X . Then each isometry $Y \rightarrow K$ can be extended to an isometry $X \rightarrow K$.

PROOF. It suffices (Zorn's lemma) to show that an isometry $f: Y \rightarrow K$ can be extended to an isometry $\bar{f}: Y \cup \{a\} \rightarrow K$ where $a \in X$. Further we may suppose that Y is closed and that $d(a, Y) := \inf\{d(a, y) : y \in Y\} > 0$. We distinguish two cases.

(1) a has no best approximation in Y . Then choose $x_1, x_2, \dots$ in Y such that $d(a, x_1), d(a, x_2), \dots$ is a strictly decreasing sequence converging to $d(a, Y)$. For each $n \in \mathbb{N}$ define the ball B_n in K by $B_n := \{\lambda \in K : |\lambda - f(x_n)| \leq d(a, x_n)\}$. Then $B_1 \supset B_2 \supset \dots$. (Indeed, if $z \in B_{n+1}$ then $|z - f(x_{n+1})| \leq d(a, x_{n+1})$, so

$$\begin{aligned} |z - f(x_n)| &\leq \max(|z - f(x_{n+1})|, |f(x_{n+1}) - f(x_n)|) \leq \\ &\max(d(a, x_{n+1}), d(x_{n+1}, x_n)) \leq \max(d(a, x_{n+1}), d(x_{n+1}, a), d(a, x_n)) = d(a, x_n) \end{aligned}$$

which proves that $z \in B_n$.) By spherical completeness the intersection is not empty; define $\bar{f}(a)$ to be arbitrary element of $\bigcap_n B_n$. To prove that $\bar{f}$ is an isometry it suffices to show that $|\bar{f}(a) - f(x)| = d(a, x)$ for each $x \in Y$. Since x is not a best approximation of a in Y there is an $n \in \mathbb{N}$ such that $d(a, x_n) < d(a, x)$ so that $d(a, x) = d(x, x_n) = |f(x) - f(x_n)|$. As $\bar{f}(a) \in B_n$ we have $|\bar{f}(a) - f(x_n)| \leq d(a, x_n) < d(a, x) = |f(x) - f(x_n)|$. Hence,

$$|\bar{f}(a) - f(x)| = \max(|\bar{f}(a) - f(x_n)|, |f(x_n) - f(x)|) = |f(x_n) - f(x)| = d(a, x)$$

and we are done.

(2) a has best approximations in Y . Let A_1 be the set of these approximations i.e. $A_1 = \{x \in Y : d(a, x) = d(a, Y)\}$. Observe that $x, y \in A_1$ implies $d(x, y) \leq d(a, Y)$. Let A_2 be a maximal subset of A_1 with the property: "if $x, y \in A_2$, $x \neq y$, then $d(x, y) = d(a, Y)$ ". The points of $A_2 \cup \{a\}$ are equidistant so by (ii) the cardinality of A_2 is strictly less than the cardinality of k . On the other hand, the points of $f(A_2)$ are contained in a ball in K of the form $B := \{\beta \in K : |\beta - \alpha| \leq d(a, Y)\}$. The equivalence relation $\sim$ defined on B by ' $\beta \sim \beta'$ if $|\beta - \beta'| < d(a, Y)$ ' yields a quotient $B/\sim$ that has the cardinality of k (by (i)). Thus the map $A_2 \xrightarrow{f} B \rightarrow B/\sim$ is injective but not surjective. It follows that we can find a $\gamma \in B$ such that $d(\gamma, f(m)) = d(a, m)$ for all $m \in A_2$. Set $\bar{f}(a) := \gamma$. Then $|\bar{f}(a) - f(x)| = d(a, x)$ for all $x \in A_2$. Now let $x \in A_1$. By maximality there is an $m \in A_2$ such that $d(x, m) < d(a, m) = |\bar{f}(a) - f(m)|$ whence $|\bar{f}(a) - f(x)| = \max(|\bar{f}(a) - f(m)|, |f(m) - f(x)|) = d(a, m) = d(a, x)$. Finally, let $x \in Y \setminus A_1$. Choose any $y \in A_1$. Then $d(a, x) > d(a, y)$ so that $|f(x) - f(y)| = d(x, y) = d(a, x)$ and $|\bar{f}(a) - f(y)| = d(a, y) < d(a, x)$. It follows that $|\bar{f}(a) - f(x)| = \max(|\bar{f}(a) - f(y)|, |f(y) - f(x)|) = d(a, x)$. So we have arrived at $|\bar{f}(a) - f(x)| = d(a, x)$ for all $x \in Y$, which finishes the proof.

PROOF OF THEOREM 1. Choose ([1], Theorem 1.3, Exercise 1.J, Example 4.Y) a spherically complete field K whose value group is $(0, \infty)$ and whose residue class field has a large cardinality (for example, strictly larger than the cardinality of X). Choose $a \in X$, set $Y := \{a\}$. By the Key Lemma we can extend the map $a \mapsto 0$ to an isometry $X \rightarrow K$.

PROOF OF THEOREM 2. $(\alpha) \Rightarrow (\beta)$. Let k have n elements. Then each subset A

of K consisting of equidistant points has at most n elements. Thus, $X := K$ satisfies conditions (i) & (ii) of the Key Lemma. $(\beta) \Rightarrow (\gamma)$. Let $f: K \rightarrow K$ be an isometry. Apply (β) to $f^{-1}: f(K) \rightarrow K$. (γ) follows. Finally, we prove $\neg(\alpha) \Rightarrow \neg(\gamma)$. Let k be infinite. Then let $j \leftrightarrow B_j$ ($j \in k$) be the correspondence between the elements of k and the additive cosets of $\{x \in K : |x| < 1\}$ in $\{x \in K : |x| \leq 1\}$. There is bijection $\tau: k \rightarrow k^\times (= \{x \in k, x \neq 0\})$. For each $j \in k$ let σ_j be an isometry (translation) of B_j onto $B_{\tau(j)}$. Define $\sigma: K \rightarrow K$ as follows

$$\sigma(x) := \begin{cases} x & \text{if } x \in K, |x| > 1 \\ \sigma_j(x) & \text{if } j \in k, x \in B_j. \end{cases}$$

It is easy to see that σ is an isometry $K \rightarrow \{x \in K : |x| \geq 1\}$. Let K be not spherically complete. Then there is a nested sequence $B_1 \supsetneq B_2 \supsetneq \dots$ of balls whose intersection is empty. For each $n \in \mathbb{N}$ choose $a_n \in B_n \setminus B_{n+1}$ and define $\sigma: K \rightarrow K$ by

$$\sigma(x) := \begin{cases} x - a_1 & \text{if } x \in K \setminus B_1 \\ x - a_{n+1} & \text{if } n \in \mathbb{N}, x \in B_n \setminus B_{n+1} \end{cases}$$

One verifies easily that σ is an isometry $K \rightarrow K$ that does not attain the value 0.

REMARK. The following statements follow without much effort from the preceding theory.

1. A separable ultrametric space can isometrically be embedded into a separable non-archimedean complete field.
2. For each ultrametric space X there exists a discretely valued field K , a real number $c > 1$, and a map $\sigma: X \rightarrow K$ satisfying $d(x, y) \leq |\sigma(x) - \sigma(y)| \leq cd(x, y)$ ($x, y \in X$).

REFERENCE

1. Van Rooij, A.C.M. — Non-Archimedean Functional Analysis. Marcel Dekker, New York (1978).